

Electric field: a simple plan

Worked solutions for every problem in **Physics 210 Tutorial 2**: 15, 16, 25-30, 32, and 33. The numbering matches your worksheet.

1 Draw the direction before calculating

An electric field points **away from positive charge** and **toward negative charge**. It describes the force on a positive test charge. Opposite charge signs do not automatically mean opposite field directions at a chosen point.

$$\vec{F} = q_{\text{test}}\vec{E}, \quad E_{\text{point}} = \frac{k|q|}{r^2}$$

2 For a distribution, use one small piece

Replace a tiny piece of the object by a point charge dq . Find its field, keep the component that survives symmetry, then add all the pieces by integrating.

$$dE = \frac{k|dq|}{r^2}, \quad dq = \lambda ds \quad \text{or} \quad dq = \sigma dA$$

3 Use metres and coulombs

Convert lengths before substituting. A microcoulomb is one millionth of a coulomb. Keep at least a few extra digits until the final answer.

$$1 \mu\text{C} = 10^{-6} \text{ C}, \quad 1 \text{ cm} = 10^{-2} \text{ m}$$

4 Check the answer

State magnitude **and direction**. Ask whether the answer has units N/C, whether cancelling components really cancel, and whether the far-away result behaves like a point charge.

Quantity	Meaning / value
k	$8.99 \times 10^9 \text{ N m}^2/\text{C}^2$
Vacuum permittivity	$8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$
Linear density	Charge per length, in C/m
Surface density	Charge per area, in C/m^2

Conventions: air is treated as vacuum. Point-charge models are used for the two thundercloud charge concentrations. Numerical answers use three significant figures. For ring and disk results, +x means the chosen side of the axis; the field reverses on the opposite side.

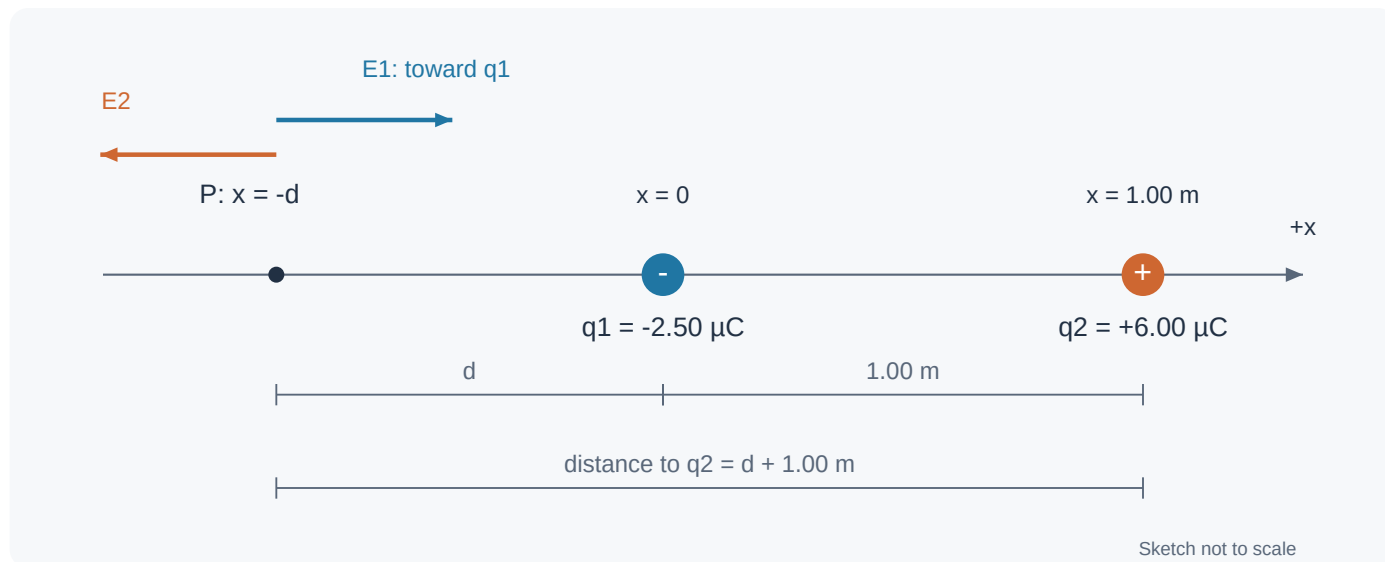
Source: user-supplied "2 E field Dist.pdf", pages 1-2. All explanatory sketches and solution steps in this document are newly prepared.

Where is the field zero?

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: $q_1 = -2.50 \mu\text{C}$ at $x = 0$; $q_2 = +6.00 \mu\text{C}$ at $x = 1.00 \text{ m}$. The separation is 1.00 m .

FIND: The finite position x where the net electric field is zero.



VARIABLES AND SYMBOLS

q_1, q_2 : The left and right source charges, measured in coulombs (C).

x ; P : x is the signed position along the horizontal axis (m); P is the point being tested.

d : Positive distance from P to q_1 when P lies to its left (m). Then $x = -d$.

$d + 1.00 \text{ m}$: Distance from P to q_2 ; it is larger because q_2 is farther right.

E_1, E_2 ; E : Fields from the two charges; E is their vector sum (N/C). At P their magnitudes must match.

Constant: $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$; $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$. $||$ means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

Where is the field zero?

Charges: **-2.50 μC** at $x = 0$ and **+6.00 μC** at $x = 1.00$ m. Find the finite point where the two fields cancel.

Use the labeled drawing and variable definitions on the preceding page.

1 Choose the correct region

Between the charges, both fields point left, so they cannot cancel. To the right of +6.00 μC , the larger positive charge is also closer, so it dominates. The zero must be **to the left of -2.50 μC** .

2 Name the distances and equate magnitudes

Let d be the positive distance left of the negative charge. The distances to the charges are d and $d + 1.00$ m. Here the fields oppose each other.

$$\frac{k(2.50 \times 10^{-6})}{d^2} = \frac{k(6.00 \times 10^{-6})}{(d + 1.00)^2}$$

3 Cancel common factors and solve

Take the positive square root because both distances are positive.

$$\frac{d + 1.00}{d} = \sqrt{\frac{6.00}{2.50}}, \quad d = \frac{1.00}{\sqrt{6.00/2.50} - 1} = 1.82085 \text{ m}$$

ANSWER

$$x = -1.82 \text{ m}$$

The point is **1.82 m left of the -2.50 μC charge**, or 2.82 m from the positive charge.

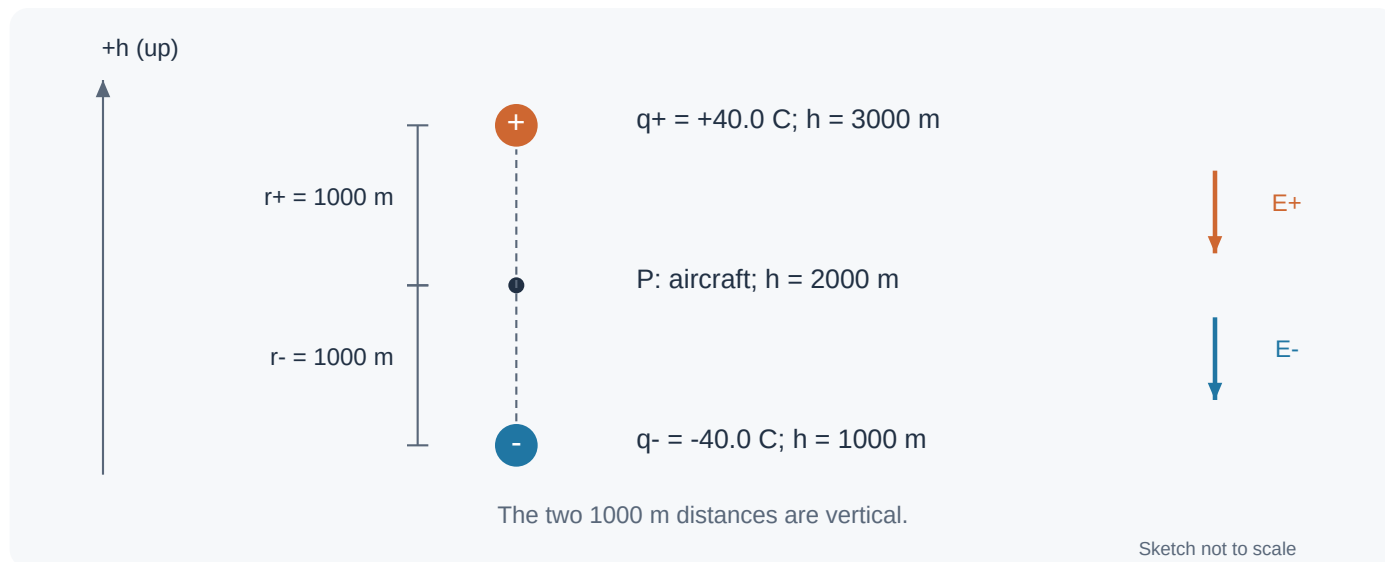
Check: the weaker charge must be closer to the zero-field point. At this point both fields have magnitude about 6.78×10^3 N/C and opposite directions.

Field at the aircraft

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: $q_+ = +40.0 \text{ C}$ at altitude 3000 m; $q_- = -40.0 \text{ C}$ at 1000 m. The aircraft is at 2000 m, directly between them.

FIND: The magnitude and direction of the net field at the aircraft.



VARIABLES AND SYMBOLS

q_+, q_- : Upper positive charge and lower negative charge (C), each treated as a point.

$P; h$: P is the aircraft position; h is altitude above the reference level (m).

r_+, r_- : Distances from P to the positive and negative charges: both 1000 m.

E_+, E_- : Fields at P produced by those charges (N/C); both point downward.

$E_{\text{one}}; E_{\text{net}}$: Magnitude from one charge; magnitude of their combined field (N/C). Upward is the positive vertical direction.

Constant: $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$; $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$. | | means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

Field at the aircraft

Aircraft altitude: 2000 m. A +40.0 C concentration is at 3000 m, and a -40.0 C concentration is at 1000 m. Treat each as a point charge.

Use the labeled drawing and variable definitions on the preceding page.

1 Find the two distances

The aircraft is 1000 m below the positive charge and 1000 m above the negative charge.

$$r_+ = r_- = 1000 \text{ m}$$

2 Determine direction before adding

The positive charge pushes a positive test charge away from itself: **downward**. The negative charge attracts it toward itself: also **downward**. Add the magnitudes.

3 Calculate one field, then double it

Both charges have the same magnitude and the same distance from the aircraft.

$$E_{\text{one}} = \frac{k(40.0)}{(1000)^2} = 3.60 \times 10^5 \text{ N/C}$$

$$E_{\text{net}} = 2E_{\text{one}}$$

ANSWER

$$E = 7.19 \times 10^5 \text{ N/C downward}$$

If upward is positive, the signed vertical component is $-7.19 \times 10^5 \text{ N/C}$.

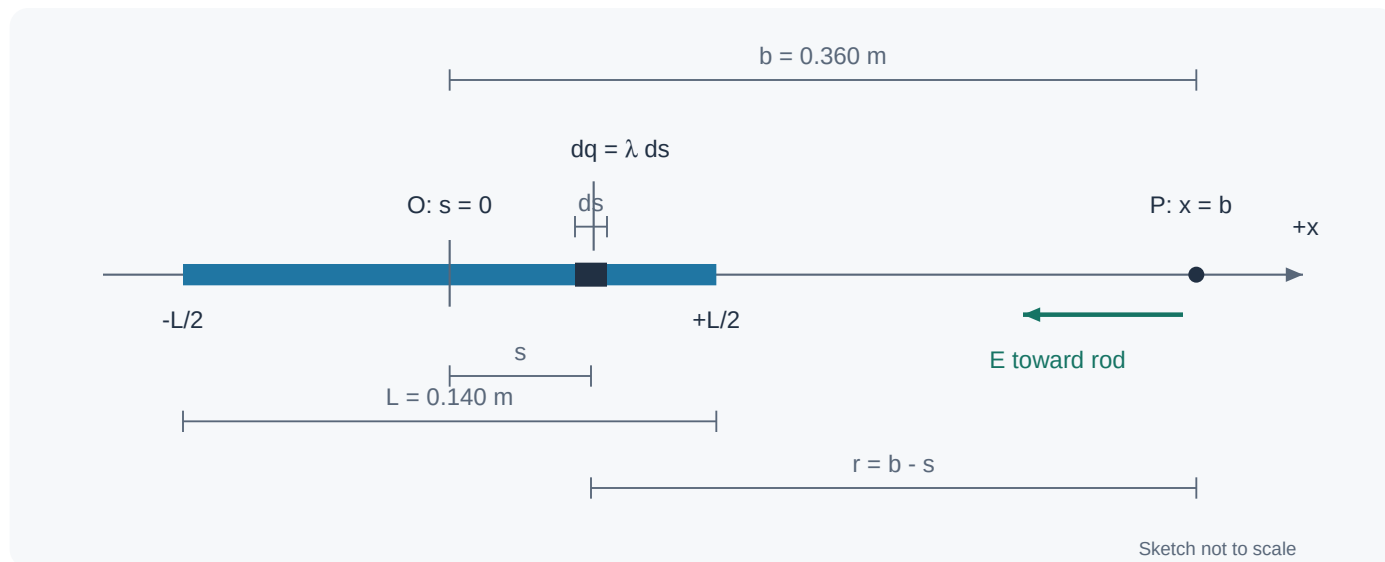
Common mistake: subtracting because the source charges have opposite signs. Direction at the observation point determines whether fields add or subtract.

Field on the axis of a rod

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: $L = 0.140$ m; $Q = -22.0$ μC ; $b = 0.360$ m from the rod center. Charge is uniform. P is drawn to the right of the rod.

FIND: The electric field magnitude and direction at P.



VARIABLES AND SYMBOLS

L; Q: Total rod length (m); total rod charge (C).

b; P: b is the distance from the center to the observation point P (m).

s; ds: s is the signed source position measured from the rod center; ds is a tiny rod length (m).

λ ; dq: $\lambda = Q/L$ is charge per length (C/m); $dq = \lambda ds$ is a tiny piece of charge (C).

$r = b - s$: Distance from a chosen source piece to P (m); it changes as s changes.

dE; E: Field magnitude of one small piece; total field magnitude after adding all pieces (N/C).

Constant: $k = 8.99 \times 10^9$ N m^2/C^2 ; $\epsilon_0 = 8.854 \times 10^{-12}$ C 2 /(N m^2). $||$ means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

Field on the axis of a rod

A uniform rod has length $L = 0.140$ m and total charge $Q = -22.0$ μC . Find the field at an axial point $b = 0.360$ m from the center. Draw the point to the right.

Use the labeled drawing and variable definitions on the preceding page.

1 Describe one small piece

Let s measure position along the rod from its center. The limits are $-L/2$ to $+L/2$. Each piece has charge magnitude $|Q| ds/L$ and is $b - s$ from the point.

$$dE = \frac{k|Q|}{L} \frac{ds}{(b-s)^2}$$

2 Add the pieces by integrating

Every contribution points toward the negative rod, so only magnitudes need integrating.

$$E = \frac{k|Q|}{L} \int_{-L/2}^{L/2} \frac{ds}{(b-s)^2} = \frac{k|Q|}{L} \left[\frac{1}{b-s} \right]_{-L/2}^{L/2}$$

3 Simplify and substitute

Subtracting the endpoint fractions gives the compact expression below.

$$E = \frac{k|Q|}{b^2 - (L/2)^2} = \frac{(8.99 \times 10^9)(22.0 \times 10^{-6})}{(0.360)^2 - (0.0700)^2}$$

ANSWER

$$E = 1.59 \times 10^6 \text{ N/C} \quad \text{toward the rod}$$

Leftward for the point shown. On the other side of the rod, the field points rightward.

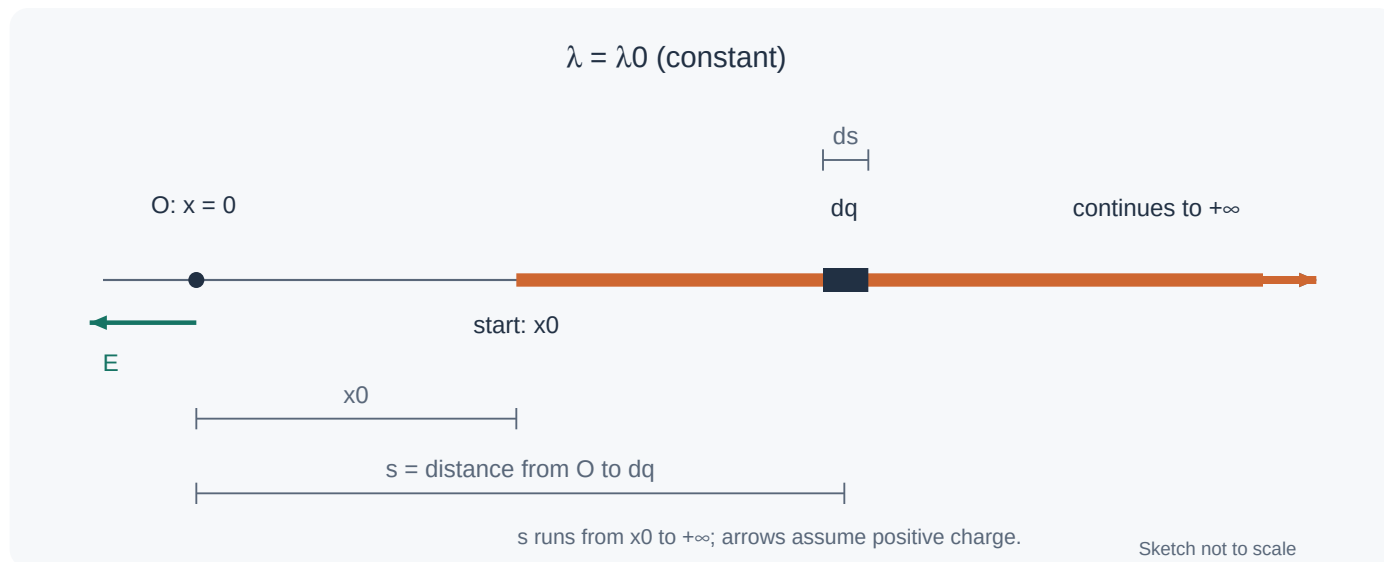
Check: the endpoint distances are 0.290 m and 0.430 m. The 0.360 m given in the question is measured from the center, not from the nearer end.

A uniform line extending to infinity

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: The line begins at $x_0 > 0$ and extends to $+\infty$. Its constant linear charge density is λ_0 . No numerical values are specified.

FIND: The electric field vector at the origin O, in terms of λ_0 and x_0 .



VARIABLES AND SYMBOLS

x_0 : Positive distance from O to the start of the charged line (m).

s; ds: s is the source coordinate and also its distance from O; ds is a tiny length (m).

λ_0 ; **dq:** Constant charge per length (C/m); $dq = \lambda_0 ds$ is a tiny charge (C).

dE_x; E_x: Tiny and total signed horizontal field components (N/C); negative means left.

E vector; |E|; x-hat: The full field, its nonnegative magnitude, and the unit vector pointing along $+x$.

Constant: $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$; $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$. $| |$ means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

A uniform line extending to infinity

The charge begins at $x = x_0 > 0$ and continues to $+\infty$. Its constant linear density is λ_0 . Find the field at the origin.

Use the labeled drawing and variable definitions on the preceding page.

1 Use the position of a piece as its distance

Call the source coordinate s . Then $dq = \lambda_0 ds$ and its distance to the origin is s . For positive charge, the field at the origin points left, along $-x$.

$$dE_x = -\frac{k\lambda_0}{s^2} ds$$

2 Integrate from the start of the line

The minus sign carries the direction. Keep the lower limit x_0 ; the line does not begin at the origin.

$$E_x = -k\lambda_0 \int_{x_0}^{\infty} s^{-2} ds = -k\lambda_0 \left[-\frac{1}{s} \right]_{x_0}^{\infty}$$

3 Evaluate the infinite endpoint

The term $1/s$ tends to zero as s tends to infinity. The remaining integral is $1/x_0$.

ANSWER

$$\vec{E} = -\frac{k\lambda_0}{x_0} \hat{\mathbf{x}}, \quad |E| = \frac{k|\lambda_0|}{x_0}$$

For $\lambda_0 > 0$, the field points left. For $\lambda_0 < 0$, it points right.

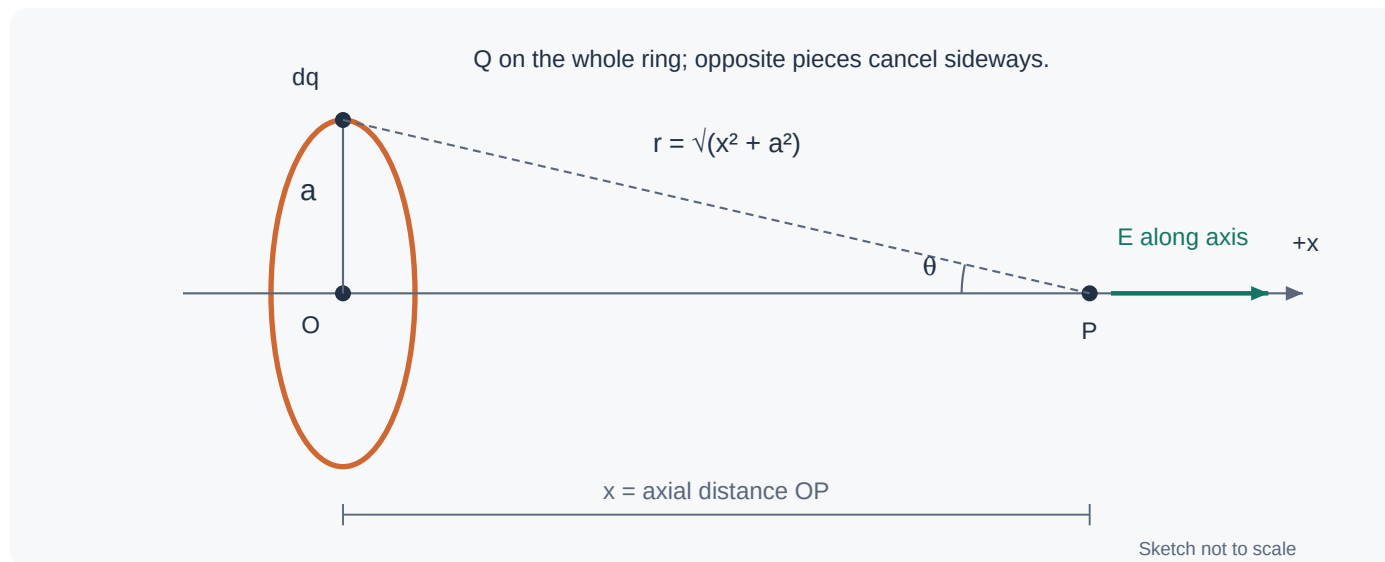
Why the answer is finite: distant pieces contribute as $1/s^2$. Although the uniform line has infinite total charge, the field integral converges.

Field on the axis of a ring

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: $a = 0.100$ m; $Q = +75.0$ μC , uniformly distributed. Axial distances x are 0.0100, 0.0500, 0.300, and 1.00 m.

FIND: The electric field magnitude and direction at each axial position.



VARIABLES AND SYMBOLS

a; Q: Ring radius (m); total ring charge (C).

O; P; x: O is the center; P is the observation point; x is the signed axial distance OP (m).

dq; r: dq is a tiny ring charge (C); $r = \sqrt{x^2 + a^2}$ is its distance to P (m).

θ ; $\cos \theta$: θ is the angle between a piece's field and $+x$. Its projection factor is $\cos \theta = x/r$.

dE; dE_x : Magnitude from one piece and its axial component: $dE_x = dE \cos \theta$ (N/C).

E_x ; E: Total signed axial field; $E = |E_x|$ is its magnitude (N/C). Transverse components cancel.

Constant: $k = 8.99 \times 10^9$ N m²/C²; $\epsilon_0 = 8.854 \times 10^{-12}$ C²/(N m²). $||$ means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

Field on the axis of a ring

Ring radius $a = 0.100$ m; total charge $Q = +75.0$ μC . Evaluate the field at the four specified distances from the center.

Use the labeled drawing and variable definitions on the preceding page.

1 Use symmetry to keep only the axial part

Opposite pieces of the ring cancel sideways. Every piece is the same distance r from the point. Its axial projection contributes a factor x/r .

$$r = \sqrt{x^2 + a^2}, \quad dE_x = \frac{k dq}{r^2} \frac{x}{r}$$

2 Add all charges and substitute

Since r and x are the same for all pieces, the integral of dq is simply Q .

$$E_x = \frac{kQx}{(x^2 + a^2)^{3/2}}$$

For example, at $x = 0.0100$ m:

$$E = \frac{(8.99 \times 10^9)(75.0 \times 10^{-6})(0.0100)}{[(0.0100)^2 + (0.100)^2]^{3/2}} = 6.64 \times 10^6 \text{ N/C}$$

Part	Distance x (m)	Field magnitude (N/C)
(a)	0.0100	6.64×10^6
(b)	0.0500	2.41×10^7
(c)	0.300	6.39×10^6
(d)	1.00	6.64×10^5

Direction for every row: away from the positively charged ring, along $+x$ for the points shown. On the opposite side, the direction reverses.

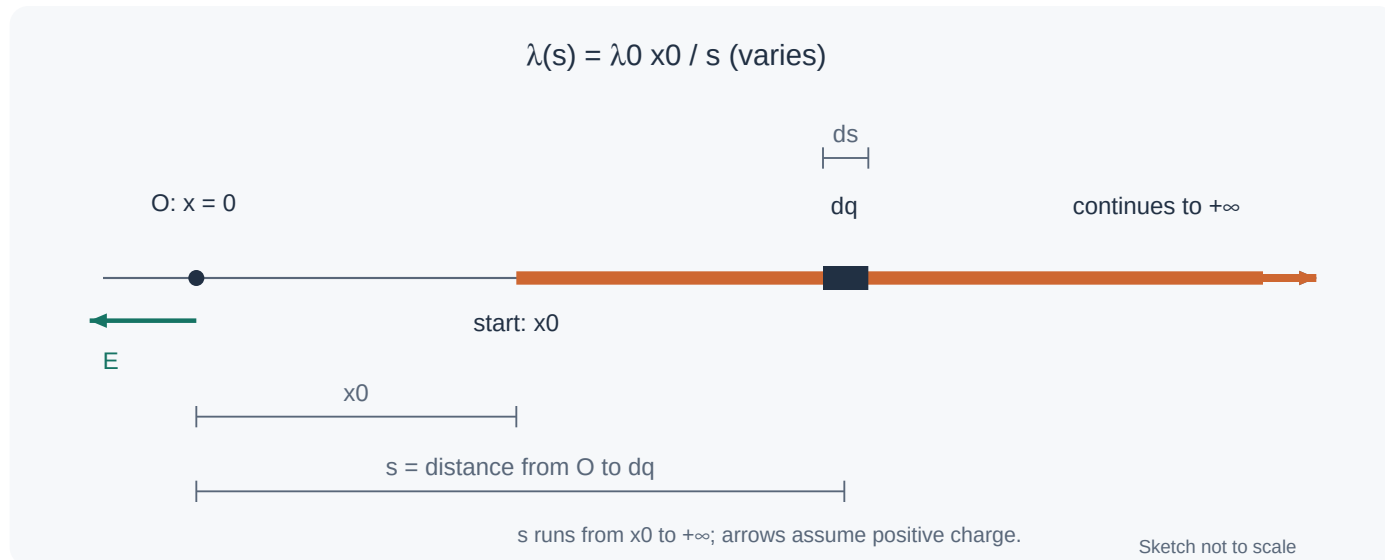
Check: the field starts at zero at the center, rises to a maximum, then decreases. It need not decrease immediately as you move away from the center.

A line with decreasing charge density

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: The line starts at $x_0 > 0$ and continues to $+\infty$. Its density varies as $\lambda(s) = \lambda_0 x_0 / s$.

FIND: The electric field vector at O, in terms of λ_0 and x_0 .



VARIABLES AND SYMBOLS

x_0 ; s : x_0 is the start position; s is the variable source position and its distance to O (m).

λ_0 ; $\lambda(s)$: Density at the start; local density at position s (C/m). Unlike Problem 26, density changes.

ds ; dq : Tiny line length (m); its charge $dq = \lambda(s) ds = (\lambda_0 x_0 / s) ds$ (C).

dE_x ; E_x : Tiny and total signed field components at O (N/C); negative means left.

E vector; $|E|$; $\hat{x}$: Full field, field magnitude, and the unit vector pointing right, along $+x$.

Constant: $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$; $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$. $| |$ means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

A line with decreasing charge density

The same line starts at $x_0 > 0$, but its density is now $\lambda(s) = \lambda_0 x_0/s$. Find the field at the origin.

Use the labeled drawing and variable definitions on the preceding page.

1 Put the varying density inside dq

The density depends on the source position s . It equals λ_0 at the start and becomes smaller farther along the line.

$$dq = \lambda(s) ds = \frac{\lambda_0 x_0}{s} ds$$

2 Include the extra distance factor

Coulomb's law contributes another $1/s^2$. Thus the field integrand is proportional to $1/s^3$, not $1/s^2$.

$$E_x = -k\lambda_0 x_0 \int_{x_0}^{\infty} s^{-3} ds$$

3 Integrate and simplify

The antiderivative is $-1/(2s^2)$; its value tends to zero at the upper limit.

$$E_x = -k\lambda_0 x_0 \left[-\frac{1}{2s^2} \right]_{x_0}^{\infty} = -\frac{k\lambda_0}{2x_0}$$

ANSWER

$$\vec{E} = -\frac{k\lambda_0}{2x_0} \hat{\mathbf{x}}, \quad |E| = \frac{k|\lambda_0|}{2x_0}$$

Leftward for positive λ_0 ; rightward for negative λ_0 .

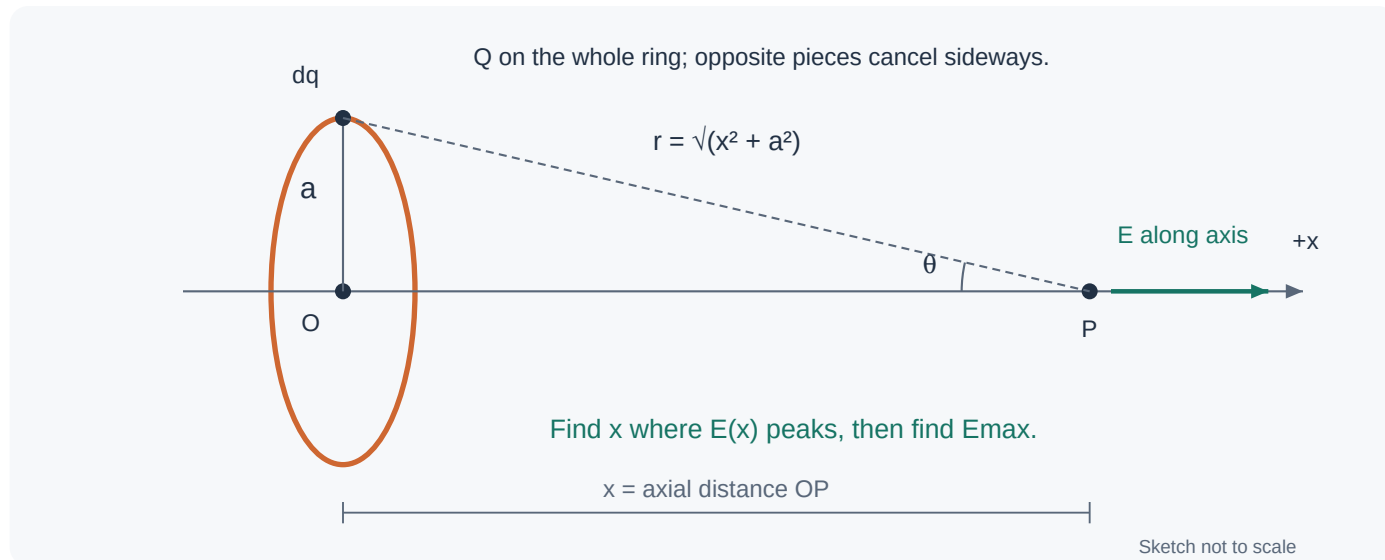
Check against Problem 26: with the same starting density, this field is exactly half the uniform-line result. The charge farther away has been reduced.

Where is the ring field largest?

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: A uniformly charged ring has radius a and nonzero total charge Q . No numerical values are specified.

FIND: The axial position(s) of maximum field magnitude and the value $E_{\max}$.



VARIABLES AND SYMBOLS

a; Q: Ring radius (m); total charge (C). Use $|Q|$ when calculating a magnitude.

x; E(x): Signed axial distance from the center (m); field magnitude on $x \geq 0$ (N/C).

O; P; r; dq; θ : Center, observation point, distance $r = \sqrt{x^2 + a^2}$ from a tiny ring charge dq to P, and angle with $\cos \theta = x/r$.

dE/dx: Rate at which field magnitude changes with position; zero at an interior maximum.

$E_{\max}$: Largest field magnitude anywhere on the axis (N/C).

ϵ_0 : Vacuum permittivity; related to Coulomb's constant by $k = 1/(4\pi\epsilon_0)$.

Constant: $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$; $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$. $| |$ means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

Where is the ring field largest?

For a ring of radius a and nonzero total charge Q , show where the axial field magnitude reaches its maximum and find that maximum.

Use the labeled drawing and variable definitions on the preceding page.

1 Work on the positive half of the axis

Use $|Q|$ for magnitude. The negative side has the same magnitude by symmetry.

$$E(x) = k|Q|x(x^2 + a^2)^{-3/2}, \quad x \geq 0$$

2 Differentiate using product and chain rules

Differentiating the x factor gives the first term; differentiating the bracket gives the second.

$$\frac{dE}{dx} = k|Q|[(x^2 + a^2)^{-3/2} - 3x^2(x^2 + a^2)^{-5/2}]$$

$$\frac{dE}{dx} = \frac{k|Q|(a^2 - 2x^2)}{(x^2 + a^2)^{5/2}}$$

3 Set the numerator to zero, then substitute

The denominator is positive. The derivative changes from positive to negative at $x = a/\sqrt{2}$, so this is a maximum.

$$a^2 - 2x^2 = 0 \quad \Rightarrow \quad x = \frac{a}{\sqrt{2}}$$

$$E_{\max} = \frac{k|Q|(a/\sqrt{2})}{(3a^2/2)^{3/2}} = \frac{2k|Q|}{3\sqrt{3}a^2}$$

ANSWER

$$|x| = \frac{a}{\sqrt{2}}, \quad E_{\max} = \frac{|Q|}{6\sqrt{3}\pi\epsilon_0 a^2}$$

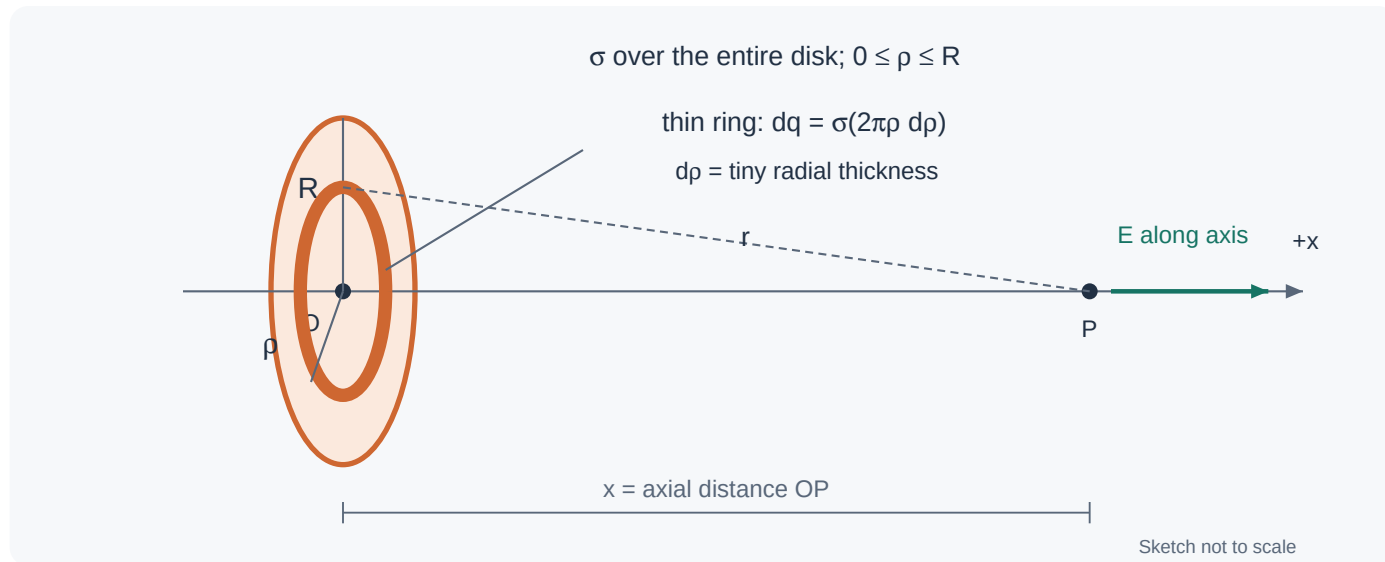
There are two symmetric maximum-magnitude locations. For positive Q on the $+x$ side, this matches the worksheet's expression.

Field on the axis of a disk

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: $R = 0.350 \text{ m}$; $\sigma = 7.90 \times 10^{-3} \text{ C/m}^2$, uniform. Axial distances x are 0.0500, 0.100, 0.500, and 2.00 m.

FIND: The electric field magnitude and direction at each axial position.



VARIABLES AND SYMBOLS

R; σ : Outer disk radius (m); charge per unit area (C/m^2).

O; P; x: Disk center, observation point, and signed axial distance OP (m).

ρ ; $d\rho$: Radius of one thin ring inside the disk and its tiny radial thickness (m); $0 \leq \rho \leq R$.

dA; dq: Thin-ring area $dA = 2\pi\rho \, d\rho$ (m^2); thin-ring charge $dq = \sigma \, dA$ (C).

r: Distance from any point on that thin ring to P: $r = \sqrt{(x^2 + \rho^2)}$ (m).

dE_x ; E_x ; E: One ring's axial field, total signed axial field, and total magnitude (N/C).

Constant: $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$; $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$. | | means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

Field on the axis of a disk

Disk radius $R = 0.350$ m; uniform surface density $\sigma = 7.90 \times 10^{-3}$ C/m². Find the axial field at the four distances.

Use the labeled drawing and variable definitions on the preceding page.

1 Build the disk from thin rings

A ring of radius ρ and thickness $d\rho$ has area $2\pi\rho d\rho$, so $dq = \sigma(2\pi\rho d\rho)$. Use the ring field from Problem 27 for each strip.

$$E_x = 2\pi k \sigma x \int_0^R \frac{\rho d\rho}{(x^2 + \rho^2)^{3/2}}$$

2 Integrate and use the disk formula

The antiderivative is $-1/\sqrt{(x^2 + \rho^2)}$. For $x > 0$, the center endpoint gives $1/x$.

$$E_x = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right), \quad \frac{\sigma}{2\epsilon_0} = 4.46 \times 10^8 \text{ N/C}$$

Part	Distance x (m)	Field magnitude (N/C)
(a)	0.0500	3.83×10^8
(b)	0.100	3.24×10^8
(c)	0.500	8.06×10^7
(d)	2.00	6.68×10^6

Sample substitution for (a): multiply 4.46×10^8 by $[1 - 0.0500/\sqrt{(0.0500^2 + 0.350^2)}] = 0.8586$.

Direction for every row: away from the positive disk, along +x on the shown side. The field reverses on the other side.

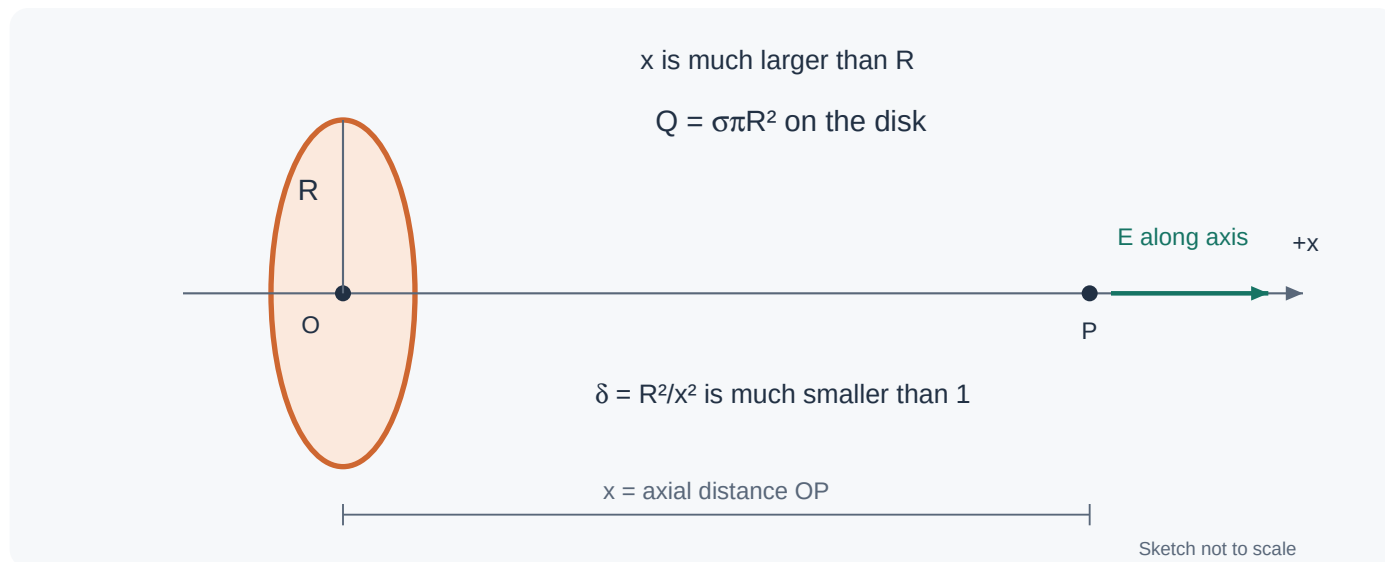
Check: just above the disk center, the field approaches $\sigma/(2\epsilon_0)$. Farther away, the disk looks like a point charge.

Why a distant disk acts like a point

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: A disk has radius R and uniform surface charge density σ . The observation point is on its axis with $x > 0$ and x much larger than R .

FIND: Show that the field approaches the point-charge expression kQ/x^2 .



VARIABLES AND SYMBOLS

R ; x : Disk radius and axial distance from its center to P (m).

σ ; Q : Surface charge density (C/m^2); total disk charge $Q = \sigma\pi R^2$ (C).

$\delta = R^2/x^2$: A dimensionless small quantity used in the binomial expansion; δ is much less than 1.

E : Signed axial field for $x > 0$ (N/C). Its magnitude is obtained using $|Q|$ or $|\sigma|$.

$\approx$; much greater than: Approximately equal to; the distance greatly exceeds the radius. The approximation improves as x/R increases.

Constant: $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$; $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$. $||$ means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

Why a distant disk acts like a point

Show that when the axial distance x is much larger than the disk radius R , the disk field approaches kQ/x^2 , where $Q = \sigma\pi R^2$. Take $x > 0$.

Use the labeled drawing and variable definitions on the preceding page.

1 Start from the exact disk result

Factor x^2 out of the square root. Because $x > 0$, $\sqrt{(x^2)} = x$.

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \left(1 + \frac{R^2}{x^2} \right)^{-1/2} \right]$$

2 Use the small quantity

If x is much larger than R , then $\delta = R^2/x^2$ is much smaller than 1. The first two binomial terms are enough for the leading field.

$$(1 + \delta)^{-1/2} \approx 1 - \frac{\delta}{2} = 1 - \frac{R^2}{2x^2}$$

3 Substitute and cancel the constant terms

The 1 in the disk expression cancels the leading 1 in the expansion. Keep the next term; it gives the nonzero field.

$$E \approx \frac{\sigma}{2\epsilon_0} \frac{R^2}{2x^2} = \frac{\sigma R^2}{4\epsilon_0 x^2}$$

4 Replace density times area by total charge

Use $Q = \sigma\pi R^2$ and $k = 1/(4\pi\epsilon_0)$.

ANSWER

$$E \approx \frac{1}{4\pi\epsilon_0} \frac{Q}{x^2} = \frac{kQ}{x^2} \quad (x \gg R)$$

This is the axial component for $x > 0$. For magnitude, replace Q with $|Q|$. Positive charge gives an outward field.

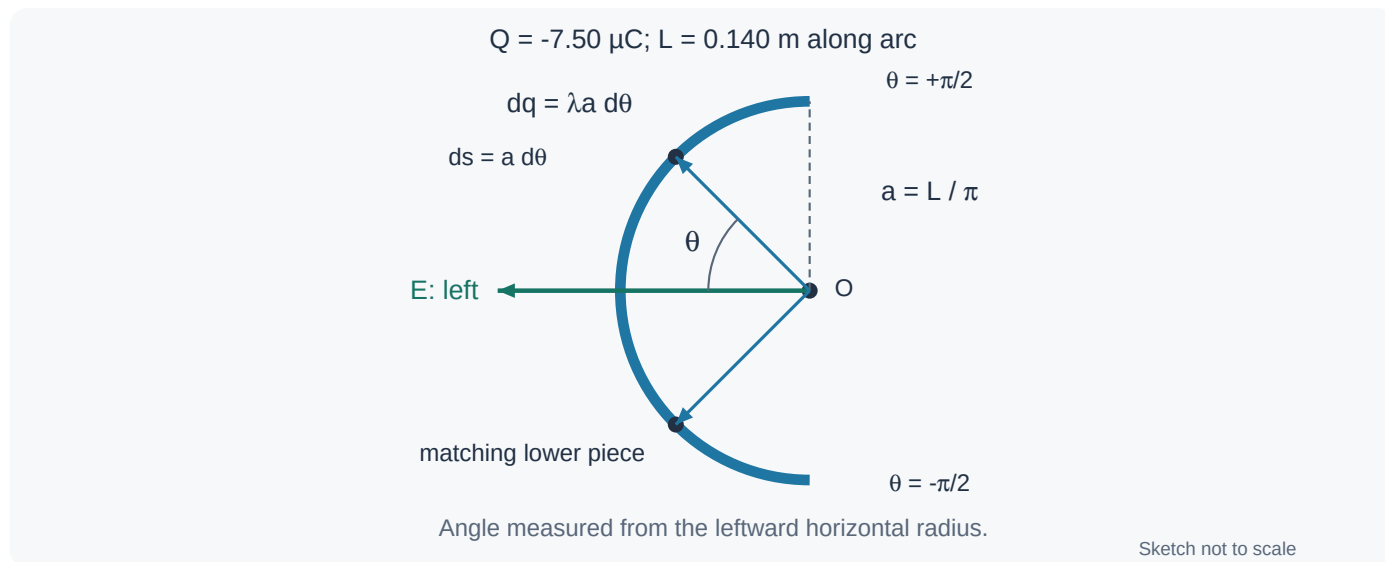
Physical meaning: from far away, the disk's size is small compared with your distance, so its detailed shape no longer matters to leading order.

Field at the center of a semicircle

SETUP / Read the known information, identify the goal, then match each symbol to the drawing.

GIVEN: $L = 0.140$ m is the curved rod's total length. $Q = -7.50$ μC is uniformly distributed on the left semicircle.

FIND: The net electric field magnitude and direction at the center O.



VARIABLES AND SYMBOLS

L; a: Arc length and semicircle radius (m), related by $L = \pi a$. L is not the radius.

Q; λ : Total charge (C); charge per length $\lambda = Q/(\pi a)$ (C/m).

θ ; $d\theta$: Angle from the leftward radius and a tiny angle step (radians); $-\pi/2 \leq \theta \leq \pi/2$.

ds; dq: Tiny arc length $ds = a d\theta$ (m); its charge $dq = \lambda a d\theta$ (C).

dE; dE_{left} : One piece's field magnitude and its leftward component $dE \cos \theta$ (N/C).

E; O: Total field magnitude (N/C); O is the center where the field is required. Vertical components cancel.

Constant: $k = 8.99 \times 10^9$ N m^2/C^2 ; $\epsilon_0 = 8.854 \times 10^{-12}$ C 2 /(N m^2). | | means magnitude; a prefix d (as in dq) denotes a tiny piece; $\int$ adds the pieces.

Continue to the next page for the full worked solution.

Field at the center of a semicircle

A rod of arc length $L = 0.140$ m is bent into the left semicircle shown. Its total charge is $Q = -7.50$ μC . Find the electric field at O.

Use the labeled drawing and variable definitions on the preceding page.

1 Convert arc length to radius

A semicircle has length πa , not $2\pi a$. Its linear charge-density magnitude is $|Q|/(\pi a)$.

$$a = \frac{L}{\pi} = 0.0446 \text{ m}, \quad |\lambda| = \frac{|Q|}{\pi a}$$

2 Cancel the vertical components

Measure θ from the leftward horizontal radius, from $-\pi/2$ to $+\pi/2$. Equal upper and lower pieces have cancelling vertical fields. Their horizontal fields both point left, toward the negative arc.

$$|dq| = |\lambda| a d\theta, \quad dE_{\text{left}} = \frac{k|\lambda|}{a} \cos \theta d\theta$$

3 Integrate the horizontal component

Every charge element is the same distance a from O.

$$E = \frac{k|\lambda|}{a} \int_{-\pi/2}^{\pi/2} \cos \theta d\theta = \frac{2k|\lambda|}{a} = \frac{2k|Q|}{\pi a^2}$$

$$E = \frac{2\pi k|Q|}{L^2} = \frac{2\pi(8.99 \times 10^9)(7.50 \times 10^{-6})}{(0.140)^2}$$

ANSWER

$$E = 2.16 \times 10^7 \text{ N/C to the left}$$

The field points from O toward the middle of the negatively charged semicircle.

Common mistakes: treating 14.0 cm as the radius, or assuming the entire field cancels. Only the vertical components cancel; the leftward components add.